



Fig. 2: CMOS gate-level netlist

a coarse-grain netlist that reduces the design to an implementation of coarse-grain cells. The netlist you obtain is simply decided by which technology library you map the design to.

Visualise how your design looks. With the tool, you can view this netlist in a gate-level representation, that is, a diagram consisting of the circuit design in terms of basic logic gates. In effect, cells that represent the corresponding function replace all operators and logic. When command Show is used for a netlist, the command invokes Graphviz, an open source software for graph visualisation. This then converts the netlist into a pictorial representation for the user to observe the design.

Interactive navigation. To analyse and investigate designs, it is often required to not just view circuit diagrams but also narrow down and scrutinise their interesting parts. Yosys allows you to change the appearance of the circuit diagram and save it in the name and format you wish. Instead of waiting for the synthesis process to complete, you can simply add command Shell at the point you want to check in the design, and synthesis is stopped at that point for you to interact with the design. Yosys also allows you to perform different types of synthesis on different sections of the code at one go.

Exchange between formats. When you talk about parsing designs and exchanging sequential logic between programs, formats like Berkeley Logic Interchange Format (BLIF) and bit-precise word-level format or BTOR come to the fore. All you need are simple commands that convert your Verilog design into these formats, and vice-versa.

More than just synthesise. Not just synthesis, Yosys lets you perform advanced analysis and circuit transformations. You can extract finite state machines and perform

Table: A default script that can be used as basis for simple synthesis scripts

```
# read design
read_verilog mydesign.v

# generic synthesis
synth -top mytop

# mapping to mycells.lib
dfflibmap -liberty mycells.lib
abc -liberty mycells.lib
clean

# write synthesised design
write_verilog synth.v
```

operations on these. It also offers minimum support to frequent sub-circuit mining and technology mapping by finding sub-circuit isomorphism.

Latest additions. Yosys 0.5 sports a log file even for warnings obtained while synthesising your code so as to make sure your design can be re-worked to perfection. Flip-flops with an enable signal promise to be a boon for digital circuit design. Commands for equivalence checking, block random access memory (BRAM) support, enhanced commands, drivers and performance improvements add a spark to Yosys.

Mapping to the technology library

One of the main aim of synthesis is to map your design to a particular technology library. To understand why this step is necessary, imagine you are drawing a circuit layout. The first thing you would need is the size or the fit-to-scale factor. In other words, the technology library provides the basic unit in the design of the layout. To make sure you can continue with your design, even if you do not have your own library, Yosys has an inbuilt technology library consisting of basic CMOS cells.

To map to the library, use command abc [options] [selection]. This invokes ABC, a software system that synthesises and verifies your logic design using scalable optimisation